Pure1™

puredrive

Name

puredrive, puredrive-admit, puredrive-list — display and manage an array's flash and NVRAM modules

Synopsis

`puredrive` admit

`puredrive` list [--csv | --nvp] [--notitle] [--page] [--raw] [--spec] [--total] [*DRIVE*...]

Arguments

DRIVE

Name of a flash module or NVRAM module about which information is to be displayed. Includes the shelf identifier (for example SH1.BAY0, which designates the module in bay #0 of storage shelf #1).

Options

-h | --help

Can be used with any command or subcommand to display a brief syntax description.

--spec

Displays hardware specifications for the module.

--total

Follows output lines with a single line containing column totals in columns where they are meaningful.

Options that control display format:

--csv

Lists information in comma-separated value (CSV) format. The `--csv` output can be used for scripting purposes and imported into spreadsheet programs.

--notitle

Lists information without column titles.

--nvp

Lists information in name-value pair (NVP) format, in the form `ITEMNAME=VALUE`. Argument names and information items are displayed flush left. The `--nvp` output is designed both for convenient viewing of what might otherwise be wide listings, and for parsing individual items for scripting purposes.

--page

Turns on interactive paging.

--raw

Displays the unformatted version of column titles and data. For example, in the `purearray monitor` output, the unformatted version of column title `us/op (read)` is `usec_per_read_op`. The `--raw` output is used to sort and filter list results.

Description

The FlashArray array consists of flash SSD (solid state drive) and NVRAM (non-volatile random access memory) modules. Flash modules are used for the persistent storage of user data, while NVRAM modules are used as non-volatile write caches. The **puredrive** command manages the flash and NVRAM modules of an array.

The **puredrive list** command lists an array's flash and NVRAM modules. The list output includes the following information:

Name

Name by which Purity//FA identifies the module in administrative operations. Flash and NVRAM module names identify physical locations in terms of shelf and bay numbers.

A bay with the name `(unknown)` means that something unexpected has happened to the bay and the array is attempting to bring it back. If the module remains in `unknown` status, contact Pure Storage Support.

Type

Drive type. Possible drive types include `SSD` (Solid State Drive) and `NVRAM` (Non-Volatile Random-Access Memory).

A dash (`-`) in the Type column represents a module that this array does not recognize. The module is initiated as a FlashArray module and is functional, but it does not belong to this array. For example, the module may have been pulled from another array and mistakenly placed into this array.

Status

Condition of the module. Possible module statuses include:

healthy

The module is functioning and responds to I/O commands.

unadmitted

The module has been added to the bay, and it is waiting to be admitted into the array. Run `puredrive admit` to admit all of the unadmitted modules, including this one. Once a module has been successfully admitted, it changes to `healthy` status. If the module returns to `unadmitted` status after the admission process, run the puredrive admit command again. If the subsequent "admit" attempt is not successful, contact Pure Storage Support.

unused

The bay does not have any modules. New modules can be added to this empty bay.

unrecognized

The module is initiated as a FlashArray module and is functional, but it does not belong to this array. For example, the module may have been pulled from another array and mistakenly placed into this array.

unhealthy

The module is not working as intended, but it may still be functional. Contact Pure Storage Support.

missing

The array is expecting a module to be in the bay, but the module is missing. Contact Pure Storage Support.

failed

The module has experienced a failure and is not responding to I/O. As a result, Purity//FA is evacuating or has evacuated data off the module. Contact Pure Storage Support.

Modules might also go into any of the following temporary statuses:

identifying

The module is in the process of coming online as Purity//FA tries to identify the drive. The module is not yet ready to respond to I/O commands. The `identifying` status is transitory and will eventually change to `unadmitted` or `healthy` status. If the module remains in `identifying` status for more than 10 minutes, contact Pure Storage Support.

recovering

The module is undergoing a drive profiling procedure or rehabilitation. The `recovering` status is transitory and will eventually change to `healthy` status. If the module remains in `recovering` status or transitions to `failed` status, contact Pure Storage Support.

updating

A firmware update is taking place. The updating status is transitory and will eventually return to `healthy` status. If the module does not return to `healthy` status, contact Pure Storage Support.

Capacity

Physical storage capacity of the module.

Evac Remaining

For a missing module, the amount of data that remains to be evacuated (reconstructed and stored on other flash modules).

Last Failure

Time at which a module became non-responsive.

Last Evac Completed

Time at which the evacuation of data from a non-responsive module completed.

Protocol

Storage protocol, such as NVMe and SAS. Must include the `--spec` option to display the Protocol column.

Capacity Upgrade and Drive Admission

Increase storage capacity by adding data packs or individual drives to a shelf.

Data packs can be added to any shelf that has open space.

Individual drives can be added to any unused slot within a shelf. The drives must be DirectFlash modules, and the shelf must already contain similar-sized DirectFlash modules. The array will not admit individual drives of other types or sizes. Add each drive to the unused slots from left to right, starting with the lowest open slot.

Before you begin the upgrade, ensure your shelf has enough open spaces to hold the new packs or individual drives.

Performing a capacity upgrade is a two-step process: first, add the modules to the array; and second, admit the newly added modules.

When a module has been added to the array, its status changes from `unused` to `identifying` as Purity works to identify the module. The module transitions to its final `unadmitted` status when it has been successfully added to the array.

Run the `puredrive list` command to verify that the newly added shelves and drives are in `unadmitted` status, indicating that the modules have been successfully connected but not yet admitted to the array.

After a module has been added (connected) to the array, it must be admitted.

Run the **puredrive admit** command to admit all modules that have been added (connected) but not yet admitted to the array.

Once a drive has been successfully admitted to the array, its module status changes from `unadmitted` to `healthy`.

Run the `puredrive list` command to verify that all of the shelves and drives are in `healthy` status, indicating that the modules have been successfully admitted and are in use by the system. This completes the drive admission process.

If issues arise during the admission process, the module status changes to `unrecognized` or `failed`, or reverts to `unadmitted`. If the module changes to `unrecognized` or `failed` status, contact Pure Storage Support. If the module returns to the `unadmitted` status, run the `puredrive admit` command again. If the subsequent “admit” attempt is not successful, contact Pure Storage Support.

Examples

Example 1

> puredrive list SH0.BAY5

Lists information for the flash module in BAY5 of shelf SH0.

Example 2

> puredrive list CH0.NVB1

Lists information for the NVRAM module in NVRAM bay 1 of shelf CH0 (does not apply to FA-400 systems).

Example 3

> puredrive admit
> puredrive list

Admits all modules that have been added (connected) to the array and are waiting to be admitted. Confirms that all modules that were in `unadmitted` status are now in `healthy` status, indicating successful admission.

See Also

n/a

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